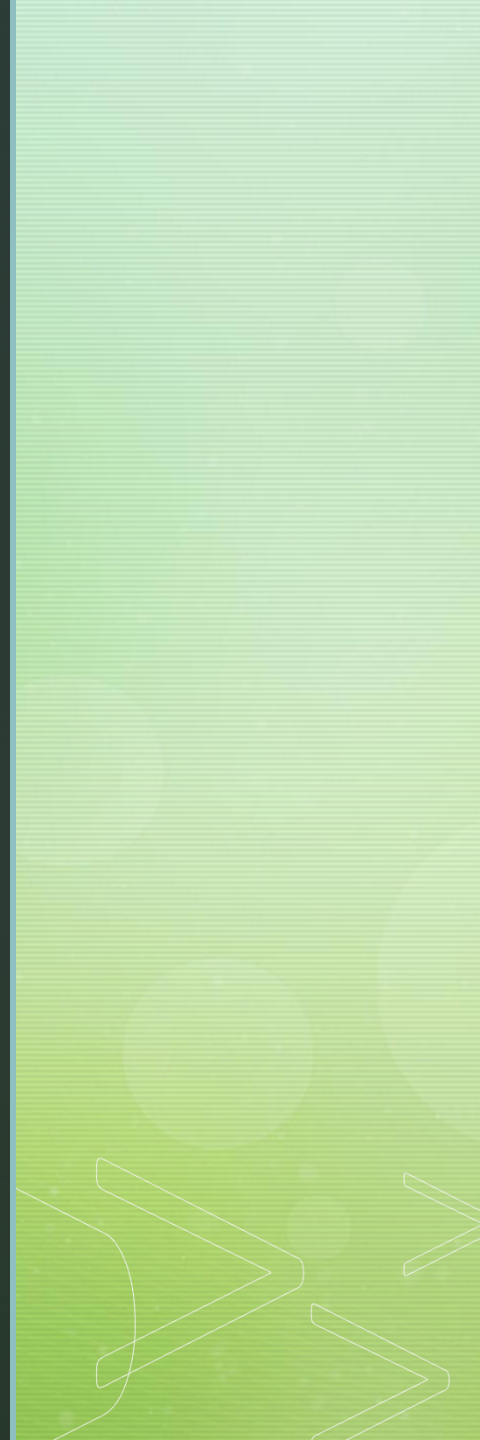




Engineering Ethics Perspective

Intellectual Property and Its Rights



Introduction to Intellectual Property (IP)

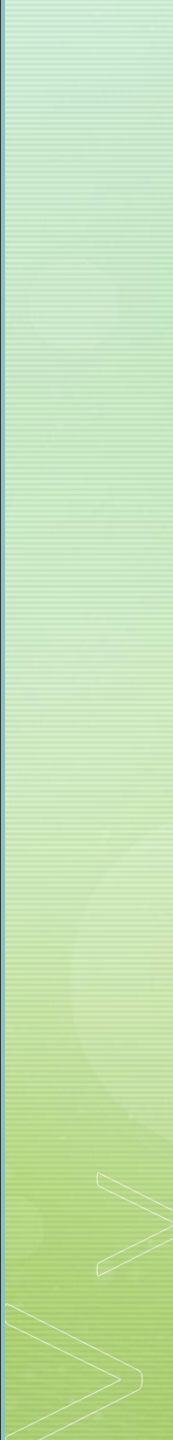
- Definition of IP: Creations of the mind, such as inventions, literary and artistic works, designs, symbols, names, and images.
- Importance for engineers and computer scientists.

Why IP Matters in Engineering Ethics

- Protects innovation and creativity.
- Prevents plagiarism and unethical practices.
- Encourages research and development.

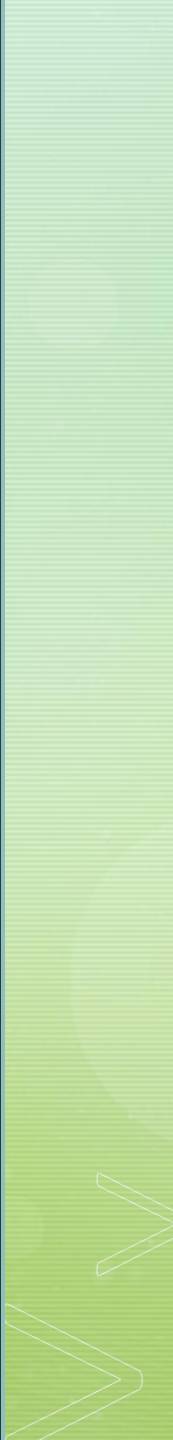


Types of Intellectual Property

- Patents
 - Copyrights
 - Trademarks
 - Trade Secrets
 - Industrial Designs
- 



Patents

- Definition: Legal protection for inventions.
 - Duration: Typically 20 years.
 - Examples: New hardware designs, innovative algorithms.
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Copyrights

- Definition: Protects literary, musical, and artistic works, including software.
- Duration: **Lifetime of author + 60 years.**
- Examples: Software code, documentation.

Trademarks

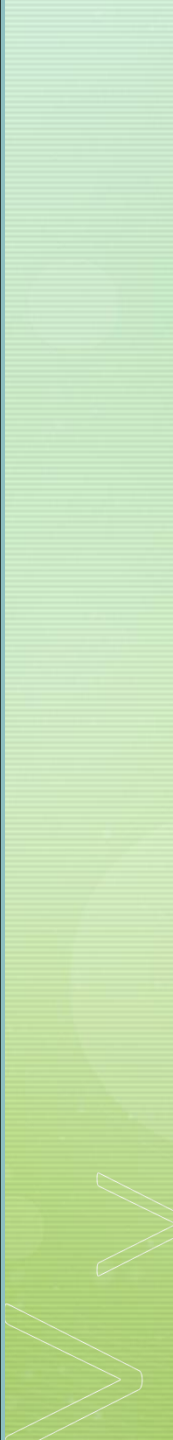
- Definition: Distinguishes goods/services through logos, names, and symbols.
- Duration: Can be renewed indefinitely.
- Examples: Tech company logos (Apple, Google).

Trade Secrets

- Definition: Confidential business information giving a competitive edge.
- Examples: Algorithms, formulas (e.g., Google PageRank).



Industrial Designs

- Definition: **Aesthetic aspects of an article.**
 - Examples: Smartphone design, UI/UX layouts.
- 

IP Rights in Software Development

- Software as **copyrightable work**.
- Open Source vs Proprietary software.
- Importance of licensing (**GPL, MIT**).



Case Study: Apple vs Samsung (Patent Dispute)

- Summary: Design patents on smartphones.
- Outcome and ethical implications.



Case Study: Oracle vs Google (Copyright)

- Issue: Use of Java APIs in Android.
- Verdict and impact on software development.

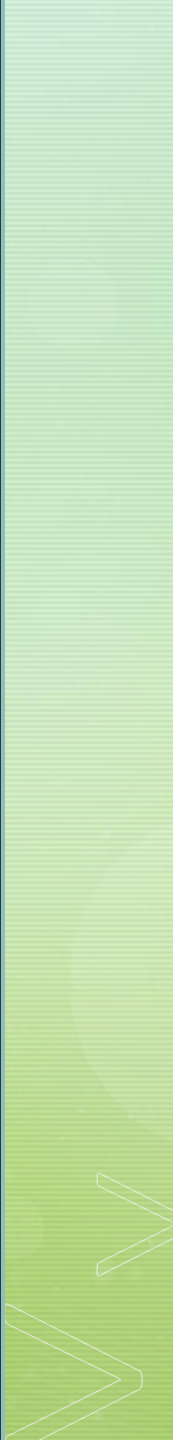


IP Violations & Consequences

- Legal penalties.
- Loss of reputation.
- Impact on careers.



Ethical Issues in IP

- Plagiarism.
 - Reverse engineering.
 - Misuse of open-source licenses.
- 



IP in Academia and Research

- Student projects and plagiarism.
- Publishing papers and copyright.
- Collaborative research challenges.

International IP Laws and Treaties

- WIPO (World Intellectual Property Organization).
- TRIPS Agreement.
- International patents.



IP and Digital Era Challenges

- Software piracy.
- Digital media sharing.
- AI-generated content issues.

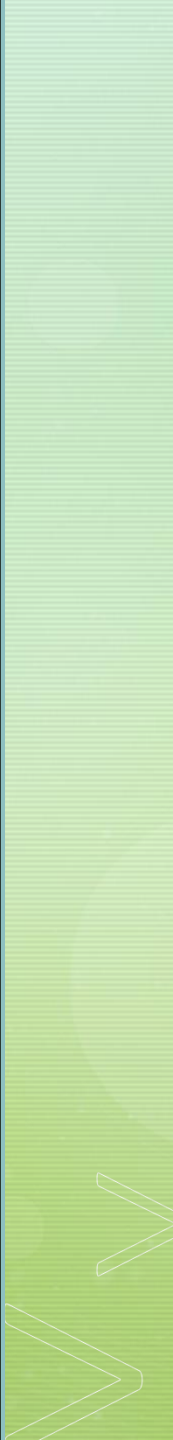


How to Protect Your IP

- Filing patents.
- Using copyright and trademarks.
- NDAs for trade secrets.



Future Trends in IP

- IP in AI and Machine Learning.
 - Blockchain for IP protection.
 - Global harmonization of IP laws.
- 



Key Takeaways

- IP encourages innovation.
 - Different forms of IP for different creations.
 - Engineers must follow IP laws and ethical practices.
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